SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA510

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I D. Naveen *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D. Naveen

50
100

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 28	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 12	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

80/100

10

Based Questions

CO1 - Assessment Task-1.

Secure Messaging System - Vigenere cipher :-

Plaintext :

HELLO

Keyword :- KEY

The vigenere cipher repeats the keyword until it matches the message length.

a. Step-by-step encryption :-

Plaintext :- H E L L O

Keyword :- K E Y K E

using :

A=0, B=1, C=2, ... Z=25.

Encryption formula :-

$$C = (P + K) \bmod 26$$

Plaintext	value	key	value	calculation	cipher
H	7	K	10	7+10 = 17	R
E	4	E	4	4+4 = 8	I
L	11	Y	24	11+24 = 35 mod 26	J
L	11	K	10	11+10 = 21	V
O	14	E	4	14+4 = 18	S

Therefore:

HELLO → RIJVS.

(b) Effect of Ku repeating keyword:

The keyword KEY is shorter than the plaintext, so it repeats

KEYKEYKEY....

For HELLO, the required key sequences is:

KEYKE

each plaintext letter is shifted according to the corresponding key letter.

Therefore, the same plaintext letter can produce different ciphertext letters depending on which keycard character is used.

∴ Final Answer:

HELLO encrypted using KEY = RIJVS.

communication security:

A student organization wants to share confidential event details with its core members. To protect the message they use a simple classical encryption technique called the Caesar cipher. In this method, each letter of the message is shifted by a fixed number of position in the alaphet.

The given message is:

Plaintext: MEET AT NOON

shifted value: 4.

a. Encryption of the message

In caesar cipher, every alaphet letter is shifted 4 positions forward.

Ex:-

M → Q

E → I

E → I

T → X

A → E

T → X

N → R

O → S

O → S

N → R

Therefore:

Meet AT NOON → QIIX EX RSSR

The spaces between the words are normaly perserved

Hence, the encrypted message is:

Ciphertext: QIIX EX RSSR

(b) Encryption of the message:-

An authorized member who knows the encryption rule and the shift value can reverse the original message. For decryption, each letter of the ciphertext is shifted 4 positions backward

Ex:

Q → M
I → E
I → E
X → T

X → A
X → T

R → N
S → O
S → O
R → N

Thus:-

QIIX EX RSSR → MEET AT NOON

Therefore, the authorized member can correctly retrieve the original confidential message.

Conclusion:-

The caesar cipher is a simple substitution encryption technique in which each letter is shifted by a fixed no. of positions In this scenario, a shift of 4 converts "meet AT NOON" into QIIX EX RSSR". By shifting the letters 4 positions backward, the original message can be recoverd.